Energy Saving Scheduling of A Single Machine System Based on Bi-objective Particle Swarm Optimization*

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***Abstract*—Energy cost becomes an important performance indicator of a production process. A great deal of energy is wasted because of the idle period resulted from the unreasonable schedule of jobs and machines. Energy consumption and other performance indicators should be combined to optimize the production schedule in shop floor. A single machine scheduling optimization problem is considered in this study, where assume that jobs are arrived at different time. An optimization model with bi-objective is formulated for minimizing the total energy consumption and the total tardiness during a shift production. Based on sequence encoding of job-machine schedule in two dimensions, a particle swarm optimization method is proposed to search Pareto solutions of the model. External archive technology is adopted to store and maintenance the non-dominated schedules. A grid density method is used to update the global best solution so far for the construction of the next generation populations. A case study is given to show the effectiveness of the algorithm based on the Pareto solutions.**

I. INTRODUCTION

The worldwide energy demand will grow by 37% till 2040, and the total carbon dioxide (CO2) emission will rise by about 80% in the later 50 years. Manufacturing industry has been the largest consumer of energy and green-house gas emitter, which accounts for 33% of the consumed energy. Manufacturing activities lead to about 38% industrial energy-related emissions of carbon dioxide. Two requirements, i.e. high efficiency and low carbon emission, are putted forwards for manufacturing systems in order to achieve sustainable development [1, 2]. Energy-aware manufacturing system gains more and more attentions all over the world [3]. A machine in idle state consumes a lot of energy. Turning off the non-bottleneck machines could save a lot of energy cost [4]. In operation research, scheduling makes a decision on when and how to allocate jobs to machines, i.e. the starting process time of each jobs on the machines. A single machine schedule is considered in this study. The motivation of the research is to schedule the machine working state with different jobs arrival time and minimize the total system energy consumption and tardiness of jobs.

The operation rules or policies can be formulated for energy efficient machine scheduling. Mouzon et al [5] presented nine switch policies for a single machine system to minimize its energy consumption. A machine was turn down if the energy consumption of one setup was less than the idle energy consumption. Their work showed that when a non-bottleneck machine was turned down, the savings of energy consumption could be 80% of no control scenario. Prabhu et al [6] proposed a discrete event simulation model for a single machine system with control policy for idle time energy consumption. The machine was switched into the low power idle state when the idle time exceeded the defined threshold. Frigerio and Matta [7] proposed a framework that integrated different policies for turning a single machine off when production was not critical and on when the job flow had to be resumed. A general policy was also formalized by modeling the energy consumption at every machine state.

Multi-objective optimization is the main method of manufacturing system scheduling. Considering energy consumption as one optimization parameter, multi-objective optimization problem is formalized and a better sequence of the jobs together with machines can be searched by optimization algorithms [8]. The scheduled product order combined with periodical shutdown for energy saving operation decreased the idle state time, which minimize the total system energy consumption and improve the energy efficiency of the manufacturing system. A mathematical model was proposed by Yildirim and Mouzon [9] to minimize total energy consumption and reduce completion time in a single machine system. A multi-objective genetic algorithm was used to get an approximate set of non-dominated schedule solutions based on dominance rules. Liu et al. [10] formulated a multi-objective optimization model in order to minimize the total CO2 emissions and completion time in a single machine system with deterministic product arrival time. A non-dominated sorting genetic algorithm II was proposed to solve the model. For a single machine scheduling problem, che et al. [11] developed a mixed-integer linear programming model with power-down mechanism to minimize total energy consumption and maximum tardiness. A basic ε-constraint method, preprocessing technique, local search, cluster analysis and valid inequalities were combined to reduce the solution space.

Energy efficient scheduling of a single machine under time-of-use (TOU) energy prices during one day was also a research direction recently. Shrouf et al. [12] proposed a mathematical model and used genetic algorithm technology to

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minimize energy consumption costs of a single machine. The job time, machine turning-on/off time and idle time were decided to avoid high-energy price periods. Fang et al. [13] presented different approximation algorithms for scheduling jobs in a single machine system with uniform-speed or scaling speed, whose aim was to minimize the electricity cost of jobs. Che et al. [14] proposed a continuous-time mixed-integer linear programming model for an energy-conscious single machine system under TOU. A greedy insertion heuristic was used to insert jobs into the available time periods one after another in non-increasing order of their electricity consumption rates. By formulating an integer programming model, an exact ε-constraint method was adopted to minimize the makespan and the total energy costs of a single machine batch scheduling problem, which have the TOU electricity prices, non-identical job sizes, and different machine energy consumption rates [15]. Cheng et al. [16] proposed a bi-objective single-machine batch scheduling problem under TOU policy, which was formulated as a mix-integer linear programming model with bi-objective, i.e. minimization of makespan and total electricity costs. For the same problem, Zhang et al. [17] derived a continuous-time mixed-integer linear programming model and two improved heuristics were proposed to guide the solution search.

As a nature-inspired evolutionary algorithm with heuristic search methods, Particle Swarm Optimization (PSO) simulates the social competitive and cooperative behavior of birds which aim to find food [18]. Many searching strategy and constraints handing methods have been proposed to improve the performances of PSO [19-21]. Due to the easy implementation and high efficiency of PSO, it has been widely used for many single-objective and multi-objective optimization problems [22,23]. In this study, a bi-objective optimization problem is formatted for a single machine system with known job arrival time considering total energy consumption and delivery tardiness of jobs . The problem is solved based on the multi-objective PSO algorithm.

II. PROBLEM STATEMENT AND OPTIMIZATION MODEL

A. Parameters and assumptions

For a single machine system, there are N independent jobs, $J_1, J_2, \ldots, J_N$, will be machined during a working day. Each job has its own arrival time, machining duration and delivery time. Considering the machining, idle and setup energy of the system, the objective of this paper is that scheduling the beginning time of each job and deciding whether or not turn off the machine when one job is finished. The overall objective is to minimize energy consumption of system and total tardiness time of jobs. The following notations are defined to describe the problem.

N: The total job quantity one day

j: The job index ($j=1,2,\ldots,N$).

R_j: The arrival time of job j.

S_j: The starting machining time of job j.

M_j: The machining duration of job j.

C_j: The ending machining time of job j.

D_j: The required delivery time of job j.

L_j: The tardiness time of job j.

$Energy_{on}$: The turning-on energy of the machine (kWh).

$Energy_{off}$: The turning-off energy of the machine (kWh).

$Time_{on}$: The turning-on time of the machine.

$Time_{off}$: The turning-off time of the machine.

$Time_{shortest}$: The required shorted time for energy saving scheduling between two jobs.

P_M: The machining power of the machine per hour (kW).

P_I: The idle power of the machine per hour (kW).

$Energy_{total}$: The total energy consumption of a working day (kWh).

$Tardiness_{total}$: The total tardiness time of the jobs.

The following assumptions are made in this study.

1) Only one job can be processed at a time on the machine.

2) The machine does not breakdown and is always available during a working day.

3) The machining process cannot be interrupted until the job is completed.

B. Optimization model of the problem

In a single machine system, the total energy consumption during a working day includes four components. The first one is turning-on energy at the beginning and turning-off energy at the end of a day. The second component is the processing energy of all jobs. The third one is the idle energy consumption during a day. The fourth part is the setup energy which is used to turn off and on the machine during a day for energy saving purpose. It is obvious that the first two energy consumptions are constant and can be described as follows:

$$Energy_{fixation} = Energy_{on} + Energy_{off} + P_M \sum_{j=1}^{n} M_j \quad (1)$$

The total idle energy and the setup energy are changed in different schedules. The idle machine should be shut down to reduce the total system energy consumption. The following two conditions must be satisfied in order to setup the machine for energy efficient system operation. The first condition is that the idle energy consumption is greater than the setup energy consumption. The second condition is the idle duration between two jobs is greater than one setup time of the machine. The required shortest time for energy saving between two jobs is defined as follows.

$$Time_{shortest} = \max\ (Time_{on} + Time_{off}, \frac{Energy_{on} + Energy_{off}}{P_I}) \quad (2)$$

The total energy consumption during a working day can be calculated as:

$$Energy_{total} = Energy_{fixation} + P_I \sum_{j=1}^{n-1} (1 - X_j)(S_{j+1} - C_j) + \sum_{j=1}^{n-1} (Energy_{on} + Energy_{off}) \times Z_j \quad (3)$$

where $j \in \{1,2,3,\ldots,N\text{-}1\}$.

$$Z_j = \begin{cases} 0, & \text{if } S_{j+1} - C_j \le Time_{shortest} \\ 1, & \text{if } S_{j+1} - C_j > Time_{shortest} \end{cases} \quad (4)$$

where Z_j=1 means the machine should be turned off when job j is finished, and Z_j=0 means the machine should keep idle or process next job immediately when job j is finished.

The tardiness time of job j is computed as:

$$L_j = C_j - D_j \quad (5)$$

When L_j is greater than the zero, the job j cannot be delivered on time and its tardiness time should be included in total tardiness time. The total tardiness time of the jobs during a working day is defined as:

$$Tardiness_{total} = \sum_{j=1}^{n} \max\left(0, L_j\right) \quad (6)$$

In order to minimize the system energy consumption and the tardiness time of jobs during a working day, the optimization model of a single machine system can be represented as:

$$\min \begin{cases} Energy_{total} \\ Tardiness_{total} \end{cases} \quad (7)$$

s.t.

$$S_j \ge R_j, \forall j = 1,2,\cdots,n \quad (8)$$

$$S_j < C_j, \forall j = 1,2,\cdots,n \quad (9)$$

$$S_{j+1} \ge C_j, \forall j = 1,2,\cdots,n-1 \quad (10)$$

$$S_j \ge 0, \forall j = 1,2,\cdots,n \quad (11)$$

$$C_j \ge 0, \forall j = 1,2,\cdots,n \quad (12)$$

In this model, Equation (8) means the starting machining time of a job is not less than its arrival time. Equation (9) ensures the starting machining time of a job is smaller than its ending machining time. Equation (10) assures the starting machining time of the latter job is greater than or equal to the previous job. Equations (11) and (12) denote that the beginning machining time and the ending machining time of each job are not less than zero.

III. PSO FOR SINGLE MACHINE SCHEDULE

In order to minimize the total energy consumption and total tardiness time, a bi-objective PSO algorithm is formulated for single machine scheduling considering deterministic job arrival time. An encoding scheme of the schedule solution is design for PSO population. An external archive and adaptive grid method are used during the evolutionary process of algorithm. The detailed procedure of the algorithm is described in this section.

Step1: Initialize population of the first generation.

1) Encoding a particle for machine schedule: Each particle is represented as a 2D vector in Table I, where variable i is the generation number of particle population. The first dimension is the job number and the second dimension is the starting processing time of the job. The starting time vector is regard as the position of a particle. We can get an N dimension particle if there are N jobs each day.

TABLE I. THE ENCODING METHOD OF A SCHEDULE

Job number	***1***	***2***	**…**	***j***	**…**	***N***
position	$x_{1,i}$	$x_{2,i}$	…	$x_{j,i}$	…	$x_{N,i}$

2) All the particles in the first population are randomly generated according constraint (8), (9) and (10). A random starting time is assigned to each job.

3) According to formula (2) and (4), the machine setup plan, i.e. vector Z, is decided for each particle. Based on the setup plan, the two objectives of each particle can be calculated by formula (3) and (6).

Step2: Initialize the external archive of non-dominated schedule solutions.

The personal best position of a particle *Pbest* and the global best position *Gbest* are determined. Store the initial non-dominated solutions into the empty archive.

Step3: Generating population of the next generation.

1) Generate a two dimensions coordinate system grid. The x axis represents the total tardiness and the y axis is the total energy consumption. The boundary of the coordinate system is the search solution space explored so far. Locate each particle in the grids based on the values of its objective functions, i.e. $Energy_{tota}$, $Tardiness_{total}$.

2) Calculate the gird index of each particle. For each grid contained more than one particle, a density value is assigned. The density value equals to the result of dividing a real number by the number of particles that the grid contains.

3) Calculate the new position of a particle based on the following formula.

$$v_{j,i+1}^k = wv_{j,i}^k + c_1 R_1 (Pbest_{j,i}^k - x_{j,i}^k) + c_2 R_2 (Gbest_{j,i}^k - x_{j,i}^k),$$

$$x_{j,i+1}^k = x_{j,i}^k + v_{j,i+1}^k \quad (13)$$

Where c_1 and c_2 are learning factors, w is the inertial weight, R_1 and R_2 are random variables generated uniformly in [0, 1].

The initial speed of a particle is set as zero. If a particle goes beyond the boundaries of a decision variable after updating its speed and position, the following equations are adopted to ensure particles flying within the search space.

$$v_{j,i+1}^k = \begin{cases} V_{\max}, & \text{if } v_{j,i+1}^k > V_{\max} \\ -v_{j,i+1}^k, & \text{if } v_{j,i+1}^k < -V_{\max} \end{cases} \quad (14)$$

$$x_{j,i+1}^k = \begin{cases} x_{j,i}^k, & \text{if } x_{j,i+1}^k > \max X_{j,i}^k \\ x_{j,i}^k, & \text{if } x_{j,i+1}^k < \min X_{j,i}^k \end{cases} \quad (15)$$

4) Determine the setup plan and calculate the two objectives of each particle. Update *Pbest* of the particle if the current solution is better than the solution in its memory. A roulette-wheel selection is used to select the target grid based on its density values. A particle is randomly selected from the target gird and set as the *Gbest*.

Step4: update the external archive.

The non-dominated schedules found at each generation are compared to the schedules in the external achieve. If a new schedule is dominated by a schedule in the external archive, a new schedule is discarded. If none of the schedules contained in the external achieve dominates a new schedule, a new schedule is stored in the external archive.

Step5: maintenance the external archive.

When the population in external archive has reached its maximum allowable capacity, some solutions should be deleted from the archive. Note that if the new inserted schedule locates outside the current bounds of the grids, the grids have to be reformulated and every schedule has to be rearranged.

Step6: output the Pareto solutions of machine schedule.

If the maximum generation is not reached, go to Setp3. Otherwise output the solutions in the external archive. The starting time of every job on machine and the setup plan of the machine are gotten.

The detailed flow of the proposed bi-objective PSO algorithm for a single machine energy saving scheduling is shown in Fig.1.

IV. CASE STUDY

For a CNC machine, the energy consumption of the normal processing is 63.4 kWh. When the machine is idle, the hydraulic system still operates normally and the energy consumption is 47.6 kWh. The turning-on time and the energy are 24 minutes and 21.3 kWh respectively. The turning-off time and required energy are 6 minutes and 1.5 kWh. According to the above data, the total setup energy of the machine per time is 22.8 kWh and the setup time is 30 minutes. It can be calculated that the required shortest duration for one setup is as follows:

$$Time_{shortest} = \max\ (30, \frac{21.3+1.5}{47.6_1} \times 60) = 30 \text{ minutes}$$

According to the optimization model, the machine should be setup if the interval time between two jobs is more than 30 minutes.

The algorithm is coded in MATLAB. The algorithm parameters in this study are as follows: c_1=1.5, c_2=1.5, w=0.5, *Size*=100, *Max*=100. The capacity of the external archive *Rep* is set as 40. Table II gives the assumption of the scheduled jobs and their arrival, processing and delivery time on this machine in one working day. For this example, the PSO algorithm can find different non-dominated schedule solution sets. Table III shows some non-dominated Pareto solutions from one randomly running of the algorithm and the corresponding Gantt charts are shown in Fig.2, where the red lines mean the delivery time of the jobs.

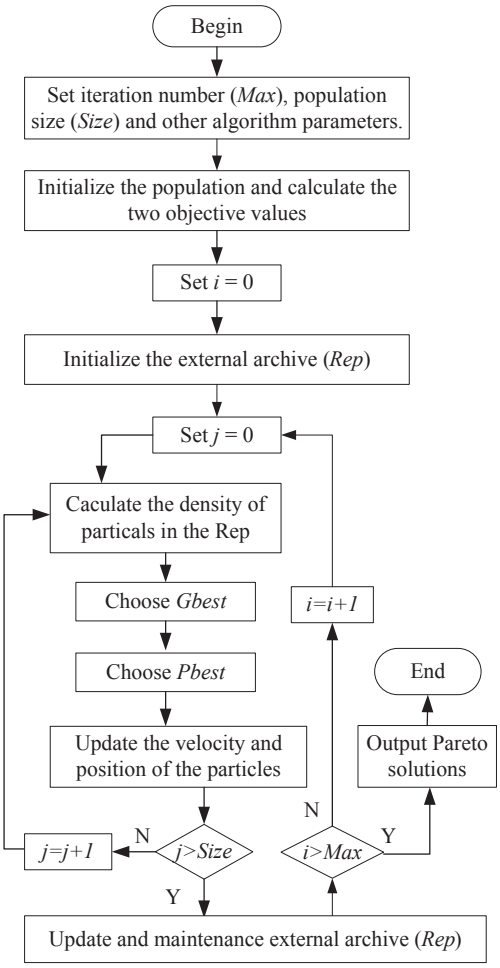


Figure 1. The flow of the bi-objective PSO algorithm for single machine

TABLE II. THE JOBS TIME OF A SINGLE MACHINE SYSTEM

Time/job	**1**	**2**	**3**	**4**	**5**	**6**	**7**
Arrival	6:00	8:00	11:30	14:30	16:00	20:00	21:00
Process	48	150	135	60	210	30	60
Delivery	8:00	12:00	16:00	18:00	22:00	00:00	00:00

TABLE III. SOME PARETO SOLUTIONS OF THE CASE EXAMPLE

	1	**2**	**3**	**4**	**5**	**6**	**7**
S1	6:00	12:00	14:30	16:45	17:45	22:15	21:15
S2	6:00	9:00	13:00	15:15	16:15	22:00	22:30
S3	6:00	9:27	15:02	17:18	18:18	22:57	21:57
S4	6:00	9:38	15:02	17:17	18:17	22:47	21:47

From Table III and Fig.2, it can be seen that solution S1 has 1 setup operations after finishing job 1. At 12:00 clocks, the machine begins to process job 2 and consciously working until all jobs are completed. This schedule has the minimum energy consumption 777.87 kWh with the total tardiness time 3.25 hours. While in solution S2, there are 3 setup operations between job 1 and job 2, job 2 and job 3, job 5 and job 6. The energy consumption and tardiness time of solution S2 are

823.47 kWh and 0 respectively, which has the minimum tardiness time. It should be note that the processing sequence of job 6 and job 7 is deferent in S1 and S2.

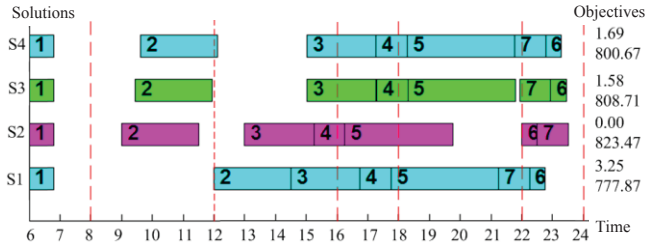


Figure 2. Gantt charts of the Pareto solutions

The system total energy consumption and jobs tardiness of some populations, i.e. the initial population, the final population, and the non-dominated Pareto solutions, are shown in Fig. 3. From the range of searched space, it can be concluded that the mathematical model and PSO algorithm are effective to find Pareto front schedules of single machine system with deterministic arrival time.

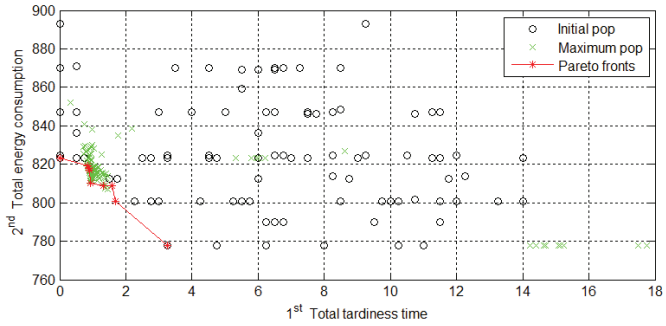


Figure 3. Some solutions of different populations and Pareto front set

V. Conclusion

With more and more attention on sustainable development, the energy cost of the idle machine cannot be ignored in manufacturing system. Turning-off the idle machine and turning-on it again at appropriate time point can be conducted based on the idle duration. For a single machine schedule problem, a bi-objective mathematical model is built considering minimization of the total energy consumption and total jobs tardiness. A PSO algorithm is proposed to search the Pareto solutions of the model, where external archive technology and grid density method is used during the evolutionary process. A case study is given to verify the algorithm and the results show its effectiveness for our bi-objective optimization problem. Considering stochastic arrival of jobs in a single machine system and the sensitivity analysis of the parameters in the algorithm are the future works. The performances comparison with other algorithms will also be investigated.

VI. References

[1] L. Zheng, P.Y. Jiang, L. H. Qiao, L. F. Xi, "Challenges and frontiers of manufacturing Systems," *Journal of Mechanical Engineering*, vol.46, no.21, pp. 124-136, 2010.

[2] C. W. Park, K. S. Kwon, W. B. Kim, et al, "Energy consumption reduction technology in manufacturing: A selective review of policies, standards, and research," *International Journal of Precision Engineering and Manufacturing*, vol.10, no.5, pp. 151-173, 2009.

[3] J.F. Wang, S. Q. Li, J. H. Liu, "A Survey on Energy Efficient Discrete Manufacturing System," *Journal of Mechanical Engineering*, vol.49, no.11, pp. 89-97, 2013.

[4] G. Mouzon, M.B. Yildirim, "A framework to minimize total energy consumption and total tardiness on a single machine," *International Journal of Sustainable Engineering*, vol.1, no.2, pp. 105-116, 2008.

[5] G. Mouzon, M. B. Yildirim, J. Twomey, "Operational methods for minimization of energy consumption of manufacturing equipment." *International Journal of Production Research,* vol.45, no.18-19, pp. 4247-4271, 2007.

[6] V. V. Prabhu, H. W. Jeon, M. Taisch, "Modeling Green Factory Physics-An Analytical Approach," *Proceedings of the IEEE Conference on Automation Science and Engineering*, Seoul, Korea, August 20-24, pp.46-51, 2012.

[7] N. Frigerio, A. Matta, "Energy-efficient control strategies for machine tools with stochastic arrivals," *IEEE Transactions on Automation Science & Engineering*, vol.12, no.1, pp. 50-61, 2014.

[8] H. Hassine, M. Barkallah, A. Bellacicco, et al, "Multi-objective optimization for sustainable manufacturing, application in turning," *International Journal of Simulation Modeling*, vol. 14, no. 1, pp.98-109, 2015.

[9] G. Mouzon, M. B. Yildirim, "Single-machine sustainable production planning to minimize total energy consumption and total completion time using a multiple objective genetic algorithm," *IEEE Transactions on Engineering Management*, vol.59, no.4, pp.585-597, 2012.

[10] C. Liu, J. Yang, J. Lian, W. Li, S. Evans, Y. Yin, "Sustainable performance oriented operational decision-making of single machine systems with deterministic product arrival time," *Journal of Cleaner Production*, vol.85, no.2, pp. 318-330, 2014.

[11] A. Che, X. Wu, J. Peng, P. Yan, "Energy-efficient bi-objective single-machine scheduling with power-down mechanism," *Computers & Operations Research*, vol.85, pp.172-183, 2017.

[12] F. Shrouf, J. Ordieres-Meré, A. García-Sánchez, et al, "Optimizing the production scheduling of a single machine to minimize total energy consumption costs," *Journal of Cleaner Production*, vol.67, no.6, pp.197-207, 2014

[13] K. Fang, N. A. Uhan, F. Zhao, J. W. Sutherland, "Scheduling on a single machine under time-of-use electricity tariffs," *Annals of Operations Research*, vol.238, no.1-2, pp. 199-227, 2016.

[14] A. Che, Y. Zeng, K. Lyu, "An efficient greedy insertion heuristic for energy-conscious single machine scheduling problem under time-of-use electricity tariffs," *Journal of Cleaner Production*, vol.129, pp. 565-577, 2016.

[15] S. Wang, M. Liu, F. Chu, C. Chu, "Bi-objective optimization of a single machine batch scheduling problem with energy cost consideration," *Journal of Cleaner Production*, vol.137, pp.1205-1215, 2016.

[16] J. Cheng, F. Chu, C. Chu, W. Xia, "Bi-objective optimization of single-machine batch scheduling under time-of-use electricity prices," *RAIRO - Operations Research*, vol.50, no.4-5, pp.715-732, 2016.

[17] S. Zhang, A. Che, X. Wu, C. Chu, "Improved mixed-integer linear programming model and heuristics for bi-objective single-machine batch scheduling with energy cost consideration," *Engineering Optimization*, vol.50, no.8, pp. 1380-1394, 2018.

[18] J. Kennedy, R. Eberhart, "Particle swarm optimization," *Proceedings of IEEE International Conference on Neural Networks*, pp.1942-1948, 1995.

[19] Q. Lin, J. Li, Z. Du, J. Chen, Z. Ming, "A novel multi-objective particle swarm optimization with multiple search strategies," *European Journal of Operational Research*, vol.247, no.3, pp.732-744, 2015.

[20] P. K. Tripathi, S. Bandyopadhyay, S. K. Pal, "An adaptive multi-objective particle swarm optimization algorithm with constraint handling," *Handbook of Swarm Intelligence*, pp. 221-239, 2011.

[21] J. D. Knowles, D. W. Corne, "Approximating the non dominated front using the pareto archived evolutionary strategy," *Evolutionary Computation*, vol.8, no.2, pp.149-172, 2000.

[22] M. Reyes-Sierra, C. C. A. Coello, "Multi-objective particle swarm optimizers: a survey of the state-of-the-art," *International Journal of Computational Intelligence Research*, no. 3, pp.287-308, 2006.

[23] D. Lei, "A pareto archive particle swarm optimization for multi-objective job shop scheduling," *Computers & Industrial Engineering*, vol.54, no.4, pp.960-971, 2008.